

AN2DL - Second Challenge Report

Backprop Boys

Alessandro Zanoni, Jacopo Valentini, Arianna Balducci, Renato Gallicola

alessandrozanoni, jacobovalentini02, ariannabalducci1, renatogallicola01

277728, 279663, 302753, 252643

December 16, 2025

1 Introduction

This project addresses *molecular subtype classification* of human tissue from whole slide images (WSI), assigning samples to four classes: **Luminal A**, **Luminal B**, **HER2(+)**, and **Triple Negative**. Accurate classification is crucial for determining appropriate treatment strategies and patient prognosis in clinical settings.

Our approach evolved through multiple iterations, starting from a whole-image classification baseline using ResNet50 [1], then transitioning to a **tile-based Multiple Instance Learning (MIL)** paradigm [2]. We ultimately developed an ensemble combining a lightweight ResNet18 tile classifier with UNI [3], a foundation model pretrained on histopathology data. The final submission achieved an F1-score of **0.4304** on the test set.

2 Problem Analysis

2.1 Dataset Characteristics

The dataset contains 1,272 WSI samples with binary masks identifying diseased tissue. After removing 60 Shrek-contaminated images, 631 clean training samples remained for the 477-image test set. The four molecular subtypes exhibit moderate class imbalance, with heterogeneous image sizes creating variability in tissue content per sample. No-

tably, several images contained green blob artifacts, whose handling is discussed in Section 3.1.

2.2 Main Challenges

Key challenges included:

1. **Limited training data:** Only 631 samples after cleaning increased overfitting risk, requiring careful regularization.
2. **Slide-level labels with tile-level features:** Each slide has a single label but contains multiple heterogeneous tissue patches, necessitating Multiple Instance Learning.
3. **Class imbalance:** Addressed through class weighting and label smoothing.
4. **Presence of Artifacts:** Green blobs and staining inconsistencies required robust pre-processing or aggregation.

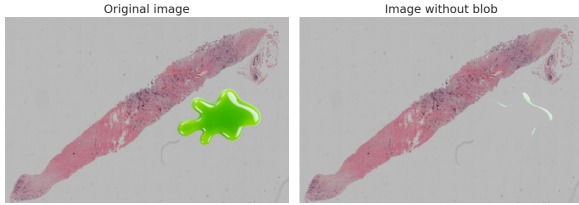
3 Method

3.1 Preprocessing Pipeline

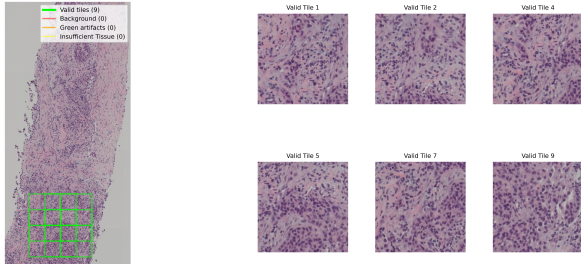
Our preprocessing strategy evolved significantly between approaches. In the initial phase, we applied aggressive blob detection using HSV color space analysis to identify and remove green artifacts. For each image, we:

1. Detected blob pixels using HSV thresholds (H: 25-100, $S \geq 80$, $V \geq 80$)
2. Estimated background color from non-tissue regions
3. Replaced blob regions with estimated background
4. Built tissue masks combining annotation masks with intensity-based detection
5. Cropped rectangular bounding boxes around tissue with 5% margin and resized to 512×512 pixels

For the tile-based approach, preprocessing was refined: 256×256 tiles were extracted using sliding windows (stride 128) from tissue regions identified by dilated masks. Each tile inherits the slide-level label, and they were filtered to exclude: (1) background regions (tissue fraction $< 20\%$), (2) green artifacts detected via HSV thresholding (H: 35-85, $S/V > 40$, green ratio $> 5\%$), and (3) tiles with minimal tumor coverage (mask fraction $< 5\%$).



(a) Blob detection and removal



(b) Tile extraction for MIL approach

3.2 Architecture Evolution

3.2.1 ResNet50 Baseline

Initial approach used ResNet50 [1] pretrained on ImageNet with frozen lower layers (layer1-3) for 3 warmup epochs, class-weighted cross-entropy with label smoothing (0.1), AdamW with cosine annealing, and standard augmentations. Limited by small

sample size and inability to capture fine-grained patterns.

3.2.2 Tile-Based ResNet18

Pivoting to MIL [2], each slide was divided into 256×256 tiles classified independently. The model was trained with flip/rotation augmentation, weighted cross-entropy (label smoothing 0.1), AdamW (lr=1e-4, wd=1e-2), 15 epochs. At inference, slide predictions obtained by averaging tile softmax probabilities, more robust than max-pooling given dataset size.

3.2.3 UNI Foundation Model

To leverage domain-specific knowledge, we incorporated UNI [3], a Vision Transformer pretrained on 100M histopathology images. We froze the UNI backbone and trained only a lightweight classifier head:

$$h = \text{UNI}_{\text{frozen}}(x), \quad y = \text{MLP}(h) \quad (1)$$

Where the MLP consists of: Linear(1024 $\rightarrow$ 256), ReLU, Dropout(0.4), Linear(256 $\rightarrow$ 4). Training: 25 epochs, bs=256, lr=1e-3, wd=1e-4.

3.3 Ensemble Strategy

3.3.1 5-Fold Cross-Validation

To optimize ensemble weights, we implemented slide-level stratified 5-fold cross-validation. For each fold: train ResNet18 on training slides' tiles and UNI classifier on training slides' cached embeddings, generate out-of-fold (OOF) slide predictions. This produced OOF probability matrices for both models across all training slides, enabling robust hyperparameter tuning without test set leakage.

3.3.2 Alpha Tuning

Ensemble via weighted averaging:

$$P_{\text{ensemble}} = \alpha P_{\text{ResNet}} + (1 - \alpha) P_{\text{UNI}} \quad (2)$$

Grid search over $\alpha \in [0, 1]$ (101 points) with local refinement yielded optimal $\alpha = 0.04$, indicating UNI's domain-specific pretraining provided stronger signal than ResNet18's ImageNet features.

3.4 Test-Time Augmentation

In the final submission, we applied Dihedral Group TTA on ResNet18 predictions. For each tile, we computed predictions under 8 transformations: 4 rotations (0° , 90° , 180° , 270°) and 4 flipped rotations (horizontal flip + 4 rotations). Softmax probabilities were averaged across all 8 views before tile-level aggregation. UNI predictions were not augmented, as preliminary experiments showed negligible gains due to its robust pretrained representations. The final slide-level prediction was:

$$P_{\text{slide}} = 0.20 \cdot \text{mean}_{\text{tiles}}(P_{\text{ResNet}}^{\text{TTA}}) + 0.80 \cdot \text{mean}_{\text{tiles}}(P_{\text{UNI}}) \quad (3)$$

Cross-validation initially suggested $\alpha = 0.04$, but this final value was based on standard inference. Since Dihedral Test-Time Augmentation was applied only to the ResNet18 stream during final testing, predictive reliability improved compared to validation. Therefore, α was empirically increased to 0.20 to reflect the stronger contribution of the TTA-augmented ResNet18 backbone.

4 Failed Experiments

Throughout the challenge, several promising approaches did not yield improvements:

- **Rotation-only TTA:** Testing with only 4 rotations (no flips) produced identical results to flip-only TTA, suggesting redundancy in the augmentation space. The dihedral group proved essential for capturing all symmetries.
- **Aggressive preprocessing:** Blob removal and tissue cropping did not improve over raw tile extraction.
- **ResNet50 whole-image approach:** Despite careful preprocessing and training strategies, the single-image paradigm struggled with the slide-to-label cardinality mismatch and limited sample size.
- **Manual alpha tuning:** Ensemble with manually selected weights underperformed compared to data-driven optimization.

5 Results and Discussion

Table 1 summarizes progressive improvements.

Table 1: F1-scores on test set. Best in **bold**.

Architecture	F1-Score
ResNet50 baseline	0.3768
ResNet18 tiles (flip TTA)	0.3859
ResNet18 + UNI ensemble	0.4020
5-fold CV + tuned α	0.4234
Dihedral TTA + ensemble	0.4304

Tile-based classification provided the foundation for success, validating MIL for WSI analysis. UNI integration yielded +16.1% relative improvement ($0.3859 \rightarrow 0.4020$), demonstrating domain-specific pretraining’s value over ImageNet features. Hyperparameter optimization through 5-fold CV further improved performance by 5.3% ($0.4020 \rightarrow 0.4234$), highlighting the importance of data-driven tuning rather than manual heuristics. Finally, Dihedral TTA contributed 1.7% ($0.4234 \rightarrow 0.4304$). The optimal $\alpha = 0.20$ heavily favors UNI (80%), though ResNet18 captures complementary patterns. Overall, this project demonstrated the effectiveness of combining lightweight convolutional architectures with large-scale foundation models for WSI classification.

6 Conclusions

Our final F1-score of **0.4304** leverages tile-based MIL, foundation models, and systematic optimization. Key insights: (1) tile-based approaches handle WSI heterogeneity effectively, (2) domain-specific pretraining (UNI) substantially outperforms ImageNet initialization, (3) CV-based hyperparameter tuning is crucial, (4) dihedral TTA exploits tissue symmetries.

Future work could explore attention-based MIL aggregators (e.g., attention-MIL, CLAM [4]) to weight informative tiles more heavily, integrate additional foundation models (e.g., CONCH, Phikon), or leverage slide-level metadata for multi-modal learning.

6.1 Contributions

A. Zanoni: ResNet baseline and ensemble coordination. **J. Valentini:** Tile extraction pipeline and MIL implementation. **A. Balducci:** UNI integration and cross-validation framework. **R. Gallicola:** Preprocessing design and TTA implementation.

References

- [1] He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep residual learning for image recognition*. CVPR, 770-778.
- [2] Ilse, M., Tomczak, J., & Welling, M. (2018). *Attention-based deep multiple instance learning*. ICML, 2127-2136.
- [3] Chen, R.J., et al. (2024). *Towards a general-purpose foundation model for computational pathology*. Nature Medicine, 30, 850-862.
- [4] Lu, M.Y., Williamson, D.F., Chen, T.Y., Chen, R.J., Barbieri, M., & Mahmood, F. (2021). *Data-efficient and weakly supervised computational pathology on whole-slide images*. Nature Biomedical Engineering, 5(6), 555-570.